

An all-exponent obstruction to the proposed joint coresets bound

Strong ℓ_p row coresets for every real $p > 2$

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Abstract

For the original-row, nonnegative-weight model in RA-05, let $S_p(k, \varepsilon)$ be the worst-case minimum support of a strong coreset. We prove that, for every fixed real $p > 2$ and all sufficiently large k ,

$$S_p(k, \varepsilon) \geq c_p \frac{k^{p/2}}{\varepsilon^{\beta_p} + (\log k)/k}, \quad 0 < \varepsilon < \frac{1}{2},$$

where $\beta_p = 2$ for positive even integers $p \geq 4$, and $\beta_p = 2 - 2/p$ otherwise. This disproves the proposed bound $\tilde{O}_p(k^{p/2}/\varepsilon + k/\varepsilon^2)$ separately for every fixed $p > 2$. For even p , it matches the cited upper bound up to logarithmic factors when $\varepsilon \geq \sqrt{(\log k)/k}$. The proof combines a rectangular, well-conditioned derivative-evaluation matrix with a sparse-mean obstruction. A finite-difference argument handles noninteger powers without assuming that their derivative kernels are positive semidefinite. We also give a self-contained explicit quartic construction and reproducible finite checks.

Scope and verification. The displayed proposed upper bound receives a complete negative answer. The optimal joint size is *not* classified here for all exponents and accuracies. The precise remaining gaps are recorded in Section 7. These are mathematical proofs prepared in this session, with an explicitly identified restricted-invertibility theorem imported from the literature. The manuscript is not peer reviewed or checked by a proof-assistant kernel. The numerical tests are supplementary, not proofs of the quantified statements.

The archive contains this PDF and its L^AT_EX source, executable verification code, machine-readable results, concrete quartic matrices and violating hyperplanes, an audit, and the earlier quartic manuscript for provenance.

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1 The exact target and the new lower bound

1.1 Model and quantifiers

For $A \in \mathbb{R}^{n \times d}$, write its rows as a_i^T . For a linear subspace F with Euclidean orthogonal projector P_F , define

$$\text{cost}_A(F) = \sum_{i=1}^n \|(I - P_F)a_i\|_2^p. \quad (1.1)$$

A strong row coreset has weights $w_i \geq 0$ such that

$$\left| \sum_i w_i \|(I - P_F)a_i\|_2^p - \text{cost}_A(F) \right| \leq \varepsilon \text{cost}_A(F) \quad (1.2)$$

for every F of dimension at most k . Its size is $|\text{supp}(w)|$, not the sum of its weights. Let $S_p(k, \varepsilon)$ be the supremum, over all n, d, A with $k < d$, of the minimum possible support in this model.

RA-05 asks for the optimal joint dependence, and in particular asks whether constants depending only on p can give

$$C_p \left(\frac{k^{p/2}}{\varepsilon} + \frac{k}{\varepsilon^2} \right) \log^{c_p} \left(\frac{2k}{\varepsilon} \right) \quad (1.3)$$

as a universal upper bound [1]. Theorem 1.2 of Lin–Mirrokni–Woodruff states an upper bound of order $\tilde{O}_p(k^{p/2}\varepsilon^{-2})$; its Section 1.4 records the joint-dependence gap [2]. We use that upper bound only to interpret our lower bounds. The lower-bound proofs do not depend on the correctness of an upper-bound algorithm.

Weights in this manuscript may depend arbitrarily on the entire input. There is no obliviousness, independence, unbiasedness, or sampling-algorithm restriction. Thus the results exclude the best possible reweighting, not merely a particular sampling scheme.

1.2 Main statements

Set

$$\beta_p = \begin{cases} 2, & p \in \{4, 6, 8, \dots\}, \\ 2 - 2/p, & p > 2 \text{ otherwise.} \end{cases} \quad (1.4)$$

All logarithms below are natural. The notation c_p, C_p denotes positive constants depending only on fixed p ; their values may differ at different occurrences.

Theorem 1.1 (All-exponent lower bound). *For each fixed real $p > 2$, there are constants $c_p > 0$ and $k_0(p)$ such that, for every integer $k \geq k_0(p)$ and every $0 < \varepsilon < 1/2$,*

$$S_p(k, \varepsilon) \geq c_p \frac{k^{p/2}}{\varepsilon^{\beta_p} + (\log k)/k}. \quad (1.5)$$

For each such k , a single input, independent of ε , gives this lower bound for all these accuracies. It has $d = k + 1$ and $n = O_p(k^{p+1})$ rows. It suffices to require preservation of hyperplane costs.

Corollary 1.2 (The proposed formula is false for every fixed exponent). *For every fixed real $p > 2$, no constants $C_p, c_p > 0$ make (1.3) a universal upper bound under the quantifiers in RA-05.*

Corollary 1.3 (Sharp moderate-accuracy order for even powers). *For each fixed even integer $p \geq 4$, the stated upper bound in [2] and Theorem 1.1 give*

$$S_p(k, \varepsilon) = \tilde{\Theta}_p(k^{p/2}\varepsilon^{-2}) \quad \text{when} \quad k \geq k_0(p), \quad \sqrt{\frac{\log k}{k}} \leq \varepsilon < \frac{1}{2}. \quad (1.6)$$

The lower bound in this regime has no additional logarithmic loss. The notation $\tilde{\Theta}_p$ allows the logarithmic factors in the cited upper bound.

Theorem 1.1 is proved in Sections 3–6. Section 7 proves the corollaries and identifies what is not determined. Section 8 supplies an independent explicit quartic obstruction, strengthening the audit trail without being needed for the all-exponent theorem.

2 Preliminaries and the one structural input

Lemma 2.1 (Hyperplane reduction). *Let $d = k + 1$. A strong coreset necessarily satisfies*

$$\left| \sum_i (w_i - 1) |a_i \cdot z|^p \right| \leq \varepsilon \sum_i |a_i \cdot z|^p \quad (z \in \mathbb{R}^d). \quad (2.1)$$

Proof. For unit z , the hyperplane $F = z^\perp$ has dimension k and $I - P_F = zz^T$. Consequently $\|(I - P_F)a_i\|_2 = |a_i \cdot z|$. Apply (1.2), then use degree- p homogeneity for arbitrary nonzero z . Both sides vanish at zero. $\square$

We use only necessity: we do not assert that preservation of hyperplanes is sufficient to preserve every smaller-dimensional subspace.

For a standard real Gaussian Z , put

$$\gamma_a = \mathbb{E}|Z|^a = \frac{2^{a/2}\Gamma((a+1)/2)}{\sqrt{\pi}}. \quad (2.2)$$

If v is uniform on the unit sphere in $\mathbb{R}^r$, $a \geq 2$, and x is a unit vector, then

$$\mathbb{E}|v \cdot x|^a \leq \gamma_a r^{-a/2}. \quad (2.3)$$

Indeed, a standard Gaussian vector is Rv , with independent radius and direction. Thus $\gamma_a = \mathbb{E}R^a \mathbb{E}|v \cdot x|^a$, and Jensen gives $\mathbb{E}R^a \geq (\mathbb{E}R^2)^{a/2} = r^{a/2}$. The formula for γ_a follows by substituting in the one-dimensional Gaussian integral.

Theorem 2.2 (Restricted invertibility; imported). *Let B be a nonzero real $d \times L$ matrix. For an integer $s \leq \text{srnk}(B) := \|B\|_F^2 / \|B\|_{\text{op}}^2$, there is a set S of s columns with*

$$\sigma_{\min}(B_S)^2 \geq \left(1 - \sqrt{\frac{s}{\text{srnk}(B)}}\right)^2 \frac{\|B\|_F^2}{L}. \quad (2.4)$$

This is the Spielman–Srivastava theorem stated as Theorem 1.1 in Marcus–Spielman–Srivastava [4]. We import the theorem in exactly this form. Its minimum singular value is the infimum over *all* coefficient vectors in $\mathbb{R}^s$. No unit-column hypothesis is required. This is the only non-elementary structural theorem imported for the principal lower bound.

Lemma 2.3 (Gaussian $2 \rightarrow p$ estimate). *If $G \in \mathbb{R}^{L \times r}$ has independent standard Gaussian entries and $p > 2$, then*

$$\mathbb{E} \|G\|_{2 \rightarrow p} \leq \sqrt{r} + \gamma_p^{1/p} L^{1/p}. \quad (2.5)$$

A proof by Gaussian interpolation is given in Appendix A. It is included to make explicit the dimension dependence that will be crucial below.

3 A core with a well-conditioned rectangular derivative matrix

Define the scalar derivative kernel

$$\psi_p(t) = t|t|^{p-2} = \frac{1}{p} \frac{d}{dt} |t|^p. \quad (3.1)$$

The next lemma does *not* assume that a matrix obtained by applying ψ_p entrywise to a Gram matrix is positive semidefinite.

Lemma 3.1 (Core lemma). *For fixed $p > 2$ and sufficiently large integer r , let*

$$L = \lfloor r^{p/2} \rfloor, \quad M = \lfloor L/48 \rfloor. \quad (3.2)$$

There are unit vectors $z_1, \dots, z_L \in \mathbb{R}^r$ and a subset $u_1, \dots, u_M$ of them such that

$$\sum_{i=1}^L |z_i \cdot x|^p \leq C_0(p) \|x\|_2^p, \quad C_0(p) = [16(1 + \gamma_p^{1/p})]^p, \quad (3.3)$$

$$E_{ij} = \psi_p(z_i \cdot u_j), \quad \sigma_{\min}(E)^2 \geq \frac{3}{64}. \quad (3.4)$$

In particular, the same moment bound holds for the u_j , and $L \leq 96M$.

Proof. Take independent standard Gaussian vectors $g_i \in \mathbb{R}^r$ and set $z_i = g_i / \|g_i\|$. Write G for the matrix with rows g_i^T . A Gaussian lower-tail bound gives

$$\mathbb{P}\{\exists i : \|g_i\| < \sqrt{r}/2\} \leq Le^{-r/8};$$

see Appendix B. For sufficiently large r , this is at most $1/8$. By Markov's inequality and Lemma 2.3, with failure probability at most $1/8$,

$$\|G\|_{2 \rightarrow p} \leq 8(\sqrt{r} + \gamma_p^{1/p} L^{1/p}) \leq 8\sqrt{r}(1 + \gamma_p^{1/p}).$$

On the intersection of these two events,

$$\sum_i |z_i \cdot x|^p \leq (2/\sqrt{r})^p \|Gx\|_p^p \leq C_0(p) \|x\|_2^p.$$

Let $K \in \mathbb{R}^{L \times L}$ have entries $K_{ij} = \psi_p(z_i \cdot z_j)$. It is real symmetric with diagonal one. For $i \neq j$, rotational invariance and (2.3) imply

$$\mathbb{E} K_{ij}^2 = \mathbb{E} |z_i \cdot z_j|^{2p-2} \leq \gamma_{2p-2} r^{-(p-1)}.$$

Therefore

$$\mathbb{E} \|K - I\|_F^2 \leq \gamma_{2p-2} L^2 r^{-(p-1)}, \quad \mathbb{P}\{\|K - I\|_F^2 > L/16\} \leq 16\gamma_{2p-2} r^{1-p/2}. \quad (3.5)$$

Because $p > 2$ is fixed, the last expression tends to zero. Increase the threshold on r so this failure probability is at most $1/8$. The three events have positive joint probability by the union bound, regardless of their dependence. Fix a realization on which all hold.

Let $\lambda_1, \dots, \lambda_L$ be the eigenvalues of K . Since $\sum_i (\lambda_i - 1)^2 \leq L/16$, at least $3L/4$ eigenvalues lie in $[1/2, 3/2]$. Let Π be the orthogonal spectral projection onto their eigenspaces and set $B = \Pi K$. Then

$$\|B\|_{\text{op}} \leq \frac{3}{2}, \quad \|B\|_F^2 \geq \frac{3L}{16}, \quad \text{srank}(B) \geq \frac{L}{12}. \quad (3.6)$$

The matrix B has L columns. Since $M \leq L/48 \leq \text{srank}(B)/4$, Theorem 2.2 selects M columns indexed by S such that

$$\sigma_{\min}(B_S)^2 \geq \frac{1}{4} \cdot \frac{3}{16} = \frac{3}{64}.$$

Choose u_j to be the corresponding z_i . Their derivative-evaluation matrix is $E = K_S$. The identity $B_S = \Pi E$ and the fact that an orthogonal projection is a contraction yield $\|Eh\| \geq \|B_S h\|$ for every $h \in \mathbb{R}^M$. Hence (3.4) holds. For $L \geq 96$, $M = \lfloor L/48 \rfloor \geq L/96$. The selected moment sum is no larger than the full one, since all summands are nonnegative. $\square$

Remark 3.2 (Why the rectangular matrix matters). The L vectors z_i are query directions. Only the selected M vectors u_j label row groups in the eventual input. Query directions need not be input rows. Spectral truncation is used solely to select columns; it does not replace a legal query by a linear combination of nonlinear costs. The inequality for $E = K_S$ remains an inequality for actual evaluations at the legal directions z_i .

4 A noise family whose sparse means cannot be too small

Lemma 4.1 (Noise lemma). *Fix $p > 2$. Define*

$$\rho_p = \frac{\gamma_p^2}{2\gamma_{2p} + (2/3)\gamma_p}, \quad D_p = \max\{1, (\log 9 + 4)/\rho_p\}, \quad (4.1)$$

$$N = \lceil D_p r^{p/2+1} \rceil, \quad q_0 = \left\lfloor \frac{r}{64 \log(17N)} \right\rfloor. \quad (4.2)$$

For sufficiently large integer r , there are unit vectors $v_1, \dots, v_N \in \mathbb{R}^r$, with mean $\mu = N^{-1} \sum_t v_t$, such that

$$\|\mu\|_2^2 \leq 8/N, \quad (4.3)$$

$$\frac{1}{N} \sum_t |v_t \cdot y|^p \leq C_1(p) r^{-p/2} \|y\|_2^p, \quad C_1(p) = 2\gamma_p(4/3)^p, \quad (4.4)$$

$$\left\| \sum_{t \in T} a_t v_t \right\|_2^2 \geq \frac{1}{64} \sum_{t \in T} a_t^2 \quad (|T| \leq q_0, \ a_t \in \mathbb{R}). \quad (4.5)$$

Moreover, $q_0 \geq c_p r / \log r \geq 1$ and $N \geq q_0$.

Proof. Choose independent uniform unit vectors v_t . We bound the failure probabilities of the three events separately.

Mean. Independence and symmetry give $\mathbb{E} \|N^{-1} \sum_t v_t\|^2 = 1/N$. Markov's inequality bounds the failure probability of (4.3) by $1/8$.

Uniform p -moment. For a fixed unit y , set $X_t = |v_t \cdot y|^p$. Then

$$0 \leq X_t \leq 1, \quad \mathbb{E} X_t \leq \gamma_p r^{-p/2}, \quad \text{Var}(X_t) \leq \gamma_{2p} r^{-p}.$$

Bernstein's inequality, in the form proved in Appendix B, implies

$$\begin{aligned} \mathbb{P} \left\{ \sum_t X_t \geq 2\gamma_p N r^{-p/2} \right\} &\leq \exp \left[-\frac{N\gamma_p^2}{2\gamma_{2p} + (2/3)\gamma_p r^{p/2}} \right] \\ &\leq \exp(-\rho_p N / r^{p/2}). \end{aligned}$$

A $1/4$ -net of the unit sphere has at most 9^r points. The union bound and the definition of D_p give failure probability at most e^{-4r} on this net. The function

$$Q(y) = \left(N^{-1} \sum_t |v_t \cdot y|^p \right)^{1/p}$$

is a seminorm. Its supremum on the unit sphere is at most $4/3$ times its maximum on the net, by the triangle inequality. Raising to the p th power proves (4.4).

Every small subset. Generate the same distribution by taking independent $g_t \sim N(0, I_r/r)$ and setting $v_t = g_t / \|g_t\|$. Write G for the matrix with columns g_t . For a fixed unit coefficient vector a supported on a fixed subset, $Ga \sim N(0, I_r/r)$, and

$$\mathbb{P}\{\|Ga\| \notin [1/2, 3/2]\} \leq 2e^{-r/16}.$$

For each set of q columns take a $1/8$ -net of its coefficient unit sphere of size at most 17^q . There are at most N^q such subsets. Union-bounding for $1 \leq q \leq q_0$ gives failure probability at most

$$2q_0 \exp(q_0 \log(17N) - r/16) \leq 2r \exp(-3r/64). \quad (4.6)$$

On the complementary event, the largest singular value of each such G_T is at most $(3/2)/(1-1/8) = 12/7 < 2$. Its smallest singular value is at least $1/2 - (1/8)(12/7) = 2/7 > 1/4$. The case $q = 1$ also bounds every column norm by 2. Normalizing the columns thus leaves smallest singular value at least $1/8$, proving (4.5).

For sufficiently large r , the sum of the three failure bounds

$$\frac{1}{8} + e^{-4r} + 2re^{-3r/64}$$

is less than one. Fix a joint realization. Finally, $N \leq 2D_p r^{p/2+1}$ for $r \geq 2$, and hence

$$\log(17N) \leq Q_p \log r, \quad Q_p = \frac{p}{2} + 1 + \frac{\log(34D_p)}{\log 2}.$$

When the unfloored quantity defining q_0 is at least two, its floor is at least half its value. Therefore $q_0 \geq r/(128Q_p \log r)$. Also $q_0 \leq r \leq N$. This proves every assertion. $\square$

5 The product input and a derivative forced by uniform accuracy

Fix the core from Lemma 3.1 and the noise from Lemma 4.1. Take the MN rows

$$a_{jt} = N^{-1/p}(u_j, v_t) \in \mathbb{R}^{2r}, \quad 1 \leq j \leq M, \quad 1 \leq t \leq N. \quad (5.1)$$

For a possibly unnormalized normal (x, y) , define

$$f(x, y) = \frac{1}{N} \sum_{j,t} |u_j \cdot x + v_t \cdot y|^p. \quad (5.2)$$

Using $|a + b|^p \leq 2^{p-1}(|a|^p + |b|^p)$ and $M \leq r^{p/2}$ gives

$$f(x, y) \leq 2^{p-1}(C_0(p) \|x\|^p + C_1(p) \|y\|^p) \leq C_*(p)(\|x\|^p + \|y\|^p), \quad (5.3)$$

where $C_*(p) = 2^{p-1} \max\{C_0(p), C_1(p)\}$.

Suppose nonnegative weights preserve every hyperplane cost with error $\varepsilon < 1/2$. Let f_w be the weighted form and $g = f_w - f$. By Lemma 2.1,

$$|g(x, y)| \leq \varepsilon f(x, y) \quad (x, y \in \mathbb{R}^r). \quad (5.4)$$

Write

$$\alpha_{jt} = w_{jt}/N, \quad s_j = \sum_t \alpha_{jt}, \quad b_j = \sum_t \alpha_{jt} v_t. \quad (5.5)$$

The original group mass is one and its original mean is μ .

5.1 Mass and noise-moment constraints

Lemma 5.1. *Under the preceding assumptions,*

$$(1 - \varepsilon)M \leq \sum_j s_j \leq (1 + \varepsilon)M, \quad (5.6)$$

$$0 \leq s_j \leq (1 + \varepsilon)C_0(p), \quad (5.7)$$

$$\sum_j s_j^2 \leq (1 + \varepsilon)^2 C_0(p)M. \quad (5.8)$$

For every unit y ,

$$\sum_{j,t} \left(\alpha_{jt} + \frac{1}{N} \right) |v_t \cdot y|^p \leq \frac{5}{2} C_1(p). \quad (5.9)$$

Proof. At $y = 0$, preservation says

$$(1 - \varepsilon) \sum_j |u_j \cdot x|^p \leq \sum_j s_j |u_j \cdot x|^p \leq (1 + \varepsilon) \sum_j |u_j \cdot x|^p.$$

Integrate over uniform unit x . Since every u_j has unit norm, all integrals of $|u_j \cdot x|^p$ equal the same strictly positive number. Cancelling it proves (5.6).

Evaluate at $x = u_j$. Nonnegativity lets us discard every term other than group j , so $s_j \leq f_w(u_j, 0) \leq (1 + \varepsilon)C_0(p)$. Multiplying the upper bound on the largest s_j by the upper bound on $\sum_j s_j$ proves (5.8).

Finally, at $x = 0$, the sum on the left of (5.9) is exactly $f_w(0, y) + f(0, y)$. Preservation bounds it by $(2 + \varepsilon)f(0, y)$. Lemma 4.1 and $M \leq r^{p/2}$ give $f(0, y) \leq C_1(p)$, proving the claim. $\square$

5.2 A global Taylor estimate that includes odd and noninteger powers

Lemma 5.2 (Uniform scalar remainder). *Let $p > 2$, $q = \lceil p \rceil - 1$, $\alpha = p - q \in (0, 1]$, and $\phi(a) = |a|^p$. Then $\phi \in C^q(\mathbb{R})$ and for all real a, b ,*

$$\left| \phi(a + b) - \sum_{\ell=0}^q \frac{\phi^{(\ell)}(a)b^\ell}{\ell!} \right| \leq 2^{1-\alpha}|b|^p \leq 2|b|^p. \quad (5.10)$$

Proof. For $t \neq 0$,

$$\phi^{(q)}(t) = (p)_q \operatorname{sign}(t)^q |t|^\alpha, \quad (p)_q = p(p-1) \cdots (p-q+1),$$

and $\phi^{(q)}(0) = 0$. The map $|t|^\alpha$ is globally α -Hölder with constant one. Its signed version is globally α -Hölder with constant at most $2^{1-\alpha}$: for opposite signs use $a^\alpha + b^\alpha \leq 2^{1-\alpha}(a+b)^\alpha$ for $a, b \geq 0$. Consequently the Hölder constant of $\phi^{(q)}$ is at most $(p)_q 2^{1-\alpha}$.

Subtract the order- q Taylor polynomial from the integral remainder of order $q-1$ to obtain

$$\frac{b^q}{(q-1)!} \int_0^1 (1-t)^{q-1} [\phi^{(q)}(a+tb) - \phi^{(q)}(a)] dt.$$

Its absolute value is at most

$$\frac{(p)_q 2^{1-\alpha} |b|^p}{(q-1)!} \int_0^1 (1-t)^{q-1} t^\alpha dt = 2^{1-\alpha} |b|^p.$$

The last equality follows from the beta integral and $(p)_q = \Gamma(p+1)/\Gamma(\alpha+1)$. This also covers integral p : then $q = p-1$ and $\alpha = 1$. $\square$

Lemma 5.3 (Exact derivative stencil). *For integer $q \geq 1$, set*

$$d_0 = -\sum_{j=1}^q \frac{1}{j}, \quad d_j = \frac{(-1)^{j+1}}{j} \binom{q}{j} \quad (1 \leq j \leq q). \quad (5.11)$$

Then $\sum_{j=0}^q d_j j^\ell$ is one for $\ell = 1$ and zero for $\ell = 0, 2, \dots, q$. Hence

$$t^{-1} \sum_{j=0}^q d_j P(jt) = P'(0)$$

for every polynomial P of degree at most q and every $t \neq 0$.

Proof. Interpolate a polynomial at the nodes $0, 1, \dots, q$ and differentiate at zero. The derivative of its j th Lagrange basis polynomial is d_j as in (5.11). Applying the interpolation identity to each monomial gives the stated moment identities. Rescaling the polynomial argument gives the formula for arbitrary t . $\square$

5.3 The uniform guarantee controls every group's mean error

Lemma 5.4 (Mixed derivative estimate). *For every pair of unit vectors x, y ,*

$$\left| \sum_j \psi_p(u_j \cdot x) \langle b_j - \mu, y \rangle \right| \leq C_p \varepsilon^{\beta_p/2}. \quad (5.12)$$

Consequently,

$$\sum_j \|b_j - \mu\|_2^2 \leq C_p M \varepsilon^{\beta_p}, \quad \sum_j \|b_j\|_2^2 \leq C_p M \varepsilon^{\beta_p} + \frac{16M}{N}. \quad (5.13)$$

Proof. For fixed unit x, y , put $h(s) = g(x, sy)$. Differentiating the finite sum gives

$$h'(0) = p \sum_j \psi_p(u_j \cdot x) \langle b_j - \mu, y \rangle. \quad (5.14)$$

First use $q = \lceil p \rceil - 1$ and the corresponding stencil. Define $\mathcal{D}_t h = t^{-1} \sum_{\ell=0}^q d_\ell h(\ell t)$ for $0 < t \leq 1$. Apply Lemma 5.2 to each atom, with scalar base $u_j \cdot x$ and increment $\ell t(v_t \cdot y)$, and cancel its Taylor terms using Lemma 5.3. In the following display the noise index is written as a to avoid confusion with the step size:

$$\begin{aligned} |\mathcal{D}_t h - h'(0)| &\leq 2t^{p-1} \left(\sum_{\ell=0}^q |d_\ell| \ell^p \right) \sum_{j,a} \left| \alpha_{ja} - \frac{1}{N} \right| |v_a \cdot y|^p \\ &\leq 5C_1(p) t^{p-1} \sum_{\ell=0}^q |d_\ell| \ell^p \leq R_p t^{p-1}. \end{aligned} \quad (5.15)$$

Here nonnegative α_{ja} gives $|\alpha_{ja} - 1/N| \leq \alpha_{ja} + 1/N$, and (5.9) bounds the sum uniformly. This is precisely where an uncontrolled total variation of signed weights would invalidate the argument.

On the other hand, (5.4) and (5.3) give

$$|\mathcal{D}_t h| \leq \frac{\varepsilon C_*(p)}{t} \sum_{\ell=0}^q |d_\ell| (1 + (\ell t)^p) \leq J_p \varepsilon / t, \quad (5.16)$$

where $J_p = C_*(p)(1 + q^p) \sum_\ell |d_\ell|$. Taking $t = \varepsilon^{1/p} \in (0, 1)$ yields

$$|h'(0)| \leq (J_p + R_p) \varepsilon^{1-1/p}. \quad (5.17)$$

If p is even, there is a sharper argument. Then h is a polynomial of degree at most p , because $|a + sb|^p = (a + sb)^p$. Use the stencil of order $q = p$ with $t = 1$. Its remainder is identically zero, and (5.4)–(5.3) give $|h'(0)| \leq C_p \varepsilon$. Together with (5.14), these two estimates prove (5.12) with (1.4).

Let H be the $M \times r$ matrix with rows $(b_j - \mu)^T$. Taking the supremum over unit y in (5.12), and then evaluating at all $x = z_i$ from the core lemma, gives

$$\|EH\|_F^2 \leq LC_p \varepsilon^{\beta_p}.$$

By (3.4), $\|EH\|_F^2 \geq (3/64) \|H\|_F^2$. Since $L \leq 96M$, the first inequality in (5.13) follows. Finally,

$$\sum_j \|b_j\|^2 \leq 2 \sum_j \|b_j - \mu\|^2 + 2M \|\mu\|^2 \leq C_p M \varepsilon^{\beta_p} + 16M/N.$$

□

6 From mean errors to support, and arbitrary ranks

For each group let $q_j = |\{t : w_{jt} \neq 0\}|$, and let $m = \sum_j q_j$ be the total support.

Lemma 6.1 (Support inequality). *With the preceding notation,*

$$m \geq c_p \frac{M}{\varepsilon^{\beta_p} + 1/q_0}. \quad (6.1)$$

Proof. If $1 \leq q_j \leq q_0$, the small-subset bound (4.5) and Cauchy–Schwarz imply

$$\|b_j\|^2 \geq \frac{1}{64} \sum_t \alpha_{jt}^2 \geq \frac{s_j^2}{64q_j}.$$

If $q_j > q_0$, nonnegativity of a squared norm gives the weaker inequality

$$\|b_j\|^2 \geq \frac{1}{64} \left(\frac{s_j^2}{q_j} - \frac{s_j^2}{q_0} \right),$$

whose right side is nonpositive. This weaker inequality therefore holds in both cases. For $q_j = 0$, put $s_j^2/q_j = 0$; then $s_j = b_j = 0$, so the same convention is consistent. Summing gives

$$\begin{aligned} \sum_j \frac{s_j^2}{q_j} &\leq 64 \sum_j \|b_j\|^2 + \frac{1}{q_0} \sum_j s_j^2 \\ &\leq C_p M (\varepsilon^{\beta_p} + 1/q_0), \end{aligned} \quad (6.2)$$

since $N \geq q_0$, (5.13) holds, and (5.8) bounds $\sum_j s_j^2$.

By (5.6), $\sum_j s_j \geq M/2 > 0$, so $m > 0$. Applying Cauchy–Schwarz over the nonempty groups,

$$(M/2)^2 \leq \left(\sum_j s_j \right)^2 \leq \left(\sum_j q_j \right) \left(\sum_j \frac{s_j^2}{q_j} \right).$$

Combine this with (6.2) to prove (6.1). □

Proof of Theorem 1.1. For dimension $2r$ and rank $2r - 1$, Lemma 6.1, $M = \Theta_p(r^{p/2})$, and $1/q_0 \leq C_p(\log r)/r$ give

$$m \geq c_p \frac{r^{p/2}}{\varepsilon^{\beta_p} + (\log r)/r}.$$

The input used no accuracy-dependent parameter. Also $MN = O_p(r^{p+1})$.

For an arbitrary sufficiently large integer k , set $r = \lfloor (k+1)/2 \rfloor$. Append $k+1-2r \in \{0, 1\}$ zero coordinates to each row of (5.1). Append the same zero coordinates to each query normal. Its orthogonal hyperplane in $\mathbb{R}^{k+1}$ has dimension exactly k , and its cost is unchanged. Thus the lower bound transfers to rank k . Since $r = \Theta(k)$ and $(\log r)/r = \Theta((\log k)/k)$, (1.5) follows. Choose $k_0(p)$ large enough for the core and noise lemmas and for $L \geq 96$. □

7 Consequences and the exact remaining scope

7.1 A polynomial contradiction for each real $p > 2$

Proof of Corollary 1.2. It suffices to use the weaker universal exponent $\beta = 2 - 2/p$, since $\beta_p \geq \beta$ and $0 < \varepsilon < 1$. Fix

$$a = \frac{\min\{p-2, 1\}}{8} > 0, \quad \varepsilon_k = k^{-a}. \quad (7.1)$$

For large k , this is an allowed accuracy. Since $a\beta < 1$, we have $(\log k)/k = o(\varepsilon_k^\beta)$. Theorem 1.1 gives a lower bound of order

$$k^{p/2+a\beta}. \quad (7.2)$$

The polynomial powers in the proposed expression are $k^{p/2+a}$ and k^{1+2a} . Their exponent gaps below (7.2) are

$$\Delta_1 = a(\beta - 1) = a \frac{p-2}{p} > 0, \quad (7.3)$$

$$\Delta_2 = \frac{p}{2} - 1 - a(2 - \beta) = \frac{p-2}{2} - \frac{2a}{p} > 0. \quad (7.4)$$

For the last inequality, $a \leq (p-2)/8$ and $p > 2$ imply $\Delta_2 > 3(p-2)/8$. Finally, $\log(2k/\varepsilon_k) = \Theta_p(\log k)$. Thus for every fixed logarithmic exponent c the ratio of the lower bound to (1.3) is bounded below by a positive constant times

$$\frac{k^{\min\{\Delta_1, \Delta_2\}}}{(\log k)^c} \rightarrow \infty.$$

No constants in (1.3) can overcome this polynomial separation. $\square$

For example, at $p = 3$ and $\varepsilon = k^{-1/8}$ our bound is $\Omega(k^{5/3})$, whereas the proposed polynomial terms are $k^{13/8}$ and $k^{5/4}$. The exponent gap from the larger term is $1/24$. At $p = 4$, taking $\varepsilon = k^{-1/4}$ and using the sharper even-power exponent gives $\Omega(k^{5/2})$, versus proposed terms $k^{9/4}$ and $k^{3/2}$.

7.2 Matching the cited upper bound, with explicit limitations

Proof of Corollary 1.3. For even p we have $\beta_p = 2$. If $\varepsilon^2 \geq (\log k)/k$, the denominator in (1.5) is at most $2\varepsilon^2$. Hence the lower bound is $c_p k^{p/2} \varepsilon^{-2}$. Theorem 1.2 in [2], with a fixed success parameter such as $\delta = 1/4$, supplies a coreset for every input with at most

$$C_p k^{p/2} \varepsilon^{-2} \left(\log \frac{4C_p k}{\varepsilon} \right)^{p+5}$$

rows. Positive success probability suffices for the existential upper bound. This proves (1.6). $\square$

For non-even p , our moderate-accuracy lower bound is of order $k^{p/2} \varepsilon^{-(2-2/p)}$ when $\varepsilon^{2-2/p} \geq (\log k)/k$. It does not match $k^{p/2} \varepsilon^{-2}$. The finite-difference proof loses this power because it balances an ε/t observation error against a t^{p-1} remainder.

For every p , the new lower bound saturates at order $k^{p/2+1}/\log k$ as ε decreases further. This is a limitation of the sparse-mean argument: it only controls noise subsets up to $q_0 = \Theta_p(k/\log k)$. The principal construction does not supply a matching all-accuracy upper bound of the same form. The

explicit quartic construction below gives a complementary lower bound at a smaller accuracy, but also does not close the full classification.

Accordingly, the exact results here are: a negative answer to the displayed proposed formula for *every* fixed $p > 2$, and a sharp order up to logarithms in the even-power regime (1.6). A status description for the broader optimal-joint-size request should still identify the non-even and smaller-accuracy gaps. No repository status has been changed by this work.

8 An independent explicit quartic counterexample

This section restates and proves the elementary counterexample from the earlier user-library manuscript [6]. Unlike the new all-exponent construction, it uses no restricted-invertibility theorem and even permits signed weights. Its finite-field construction is given in Appendix C. Real mutually unbiased bases are classical objects; see also [5].

Let

$$r = 4^h \ (h \geq 1), \quad B = r/2 + 1, \quad M = Br, \quad \lambda = 1 - 1/r. \quad (8.1)$$

Take B real orthonormal bases of $\mathbb{R}^r$ such that vectors from different bases have inner products of absolute value $r^{-1/2}$. List all their vectors as $u_1, \dots, u_M$, and let U have rows u_j^T .

Lemma 8.1 (Core identities). *Writing $G = UU^T$, with entrywise power denoted by $\circ$, we have*

$$U^T U = B I_r, \quad (8.2)$$

$$\sum_j (u_j \cdot x)^4 = \frac{3}{2} \|x\|_2^4, \quad (8.3)$$

$$K := G^{\circ 3} = (1 - 1/r) I_M + G/r \succeq \lambda I_M. \quad (8.4)$$

Proof. The first identity follows by adding the identities from each basis. For the fourth moment, index a vector as u_{bt} by basis b and coordinate t , and put $Q_{bt} = u_{bt} u_{bt}^T - I_r/r$. For distinct bases,

$$\langle Q_{bt}, Q_{cs} \rangle_F = (u_{bt} \cdot u_{cs})^2 - 1/r = 0.$$

The Q_{bt} for one basis span its $(r-1)$ -dimensional space of traceless diagonal matrices. These spaces are pairwise orthogonal and have total dimension

$$B(r-1) = r(r+1)/2 - 1,$$

which is the dimension of all traceless symmetric matrices. For a traceless symmetric X , its squared projection length onto a given space is $\sum_t \langle X, Q_{bt} \rangle_F^2$: this is the sum of squares of its diagonal entries in that basis. Therefore

$$\sum_{b,t} \langle X, Q_{bt} \rangle_F^2 = \|X\|_F^2.$$

Insert $X = xx^T - \|x\|^2 I_r/r$. Expanding gives

$$\sum_j (u_j \cdot x)^4 - \frac{B}{r} \|x\|^4 = \left(1 - \frac{1}{r}\right) \|x\|^4.$$

As $B/r = 1/2 + 1/r$, this proves (8.3). Finally, diagonal entries of G are one; off-diagonal entries are zero or $\pm r^{-1/2}$. Cubing each entry gives (8.4). Here $G \succeq 0$, as it is an ordinary Gram matrix. $\square$

Theorem 8.2 (Explicit quartic family). *Let $\delta = B^{-1/4}$ and form the Mr rows*

$$a_{jt} = r^{-1/4}(u_j, \delta e_t) \in \mathbb{R}^{2r}, \quad 1 \leq j \leq M, \quad 1 \leq t \leq r. \quad (8.5)$$

Any real weights preserving all hyperplane costs within relative error ε have support m satisfying

$$m \geq Mr \left(1 - \frac{100r\varepsilon^2}{\lambda^2\delta^2}\right). \quad (8.6)$$

In particular, for $k = 2r - 1$ and $\varepsilon_r = \delta/(20\sqrt{r})$, every strong row coreset has at least $5Mr/9 = \Omega(r^3)$ rows. This accuracy is $\Theta(r^{-3/4})$, and (1.3) at $p = 4$ is only $O(r^{11/4} \log^c r)$ for any fixed c .

Proof. For the rows (8.5), put

$$f(x, y) = \frac{1}{r} \sum_{j,t} (u_j \cdot x + \delta y_t)^4.$$

The scalar inequality $(a + b)^4 \leq 8(a^4 + b^4)$ and Lemma 8.1 imply

$$f(x, y) \leq 12 \|x\|^4 + 8 \|y\|^4, \quad (8.7)$$

since $M\delta^4/r = 1$ and $\sum_t y_t^4 \leq \|y\|^4$. Let $g = f_w - f$ and $H_{jt} = (w_{jt} - 1)/r$. The part of g odd in y is

$$O(x, y) := \frac{g(x, y) - g(x, -y)}{2} = 4\delta A(x, y) + 4\delta^3 D(x, y),$$

where $A(x, y) = \sum_{j,t} H_{jt}(u_j \cdot x)^3 y_t$ and $D(x, y) = \sum_{j,t} H_{jt}(u_j \cdot x) y_t^3$. Therefore

$$O(2x, y) - 2O(x, y) = 24\delta A(x, y). \quad (8.8)$$

For unit x, y , relative preservation and (8.7) give $|O(x, y)| \leq 20\varepsilon$ and $|O(2x, y)| \leq 200\varepsilon$. Thus $|A(x, y)| \leq 10\varepsilon/\delta$. Evaluate at $x = u_i$ and take the supremum over unit y . The result bounds every row norm of KH by $10\varepsilon/\delta$, and hence

$$\|H\|_F^2 \leq \lambda^{-2} \|KH\|_F^2 \leq \frac{100M\varepsilon^2}{\lambda^2\delta^2}.$$

Each omitted row has $H_{jt} = -1/r$. Thus $(Mr - m)/r^2 \leq \|H\|_F^2$, proving (8.6). At ε_r , $\lambda \geq 3/4$ gives

$$m \geq Mr \left(1 - \frac{1}{4\lambda^2}\right) \geq \frac{5}{9}Mr.$$

The claimed polynomial comparison follows from $k = \Theta(r)$ and $\varepsilon_r = \Theta(r^{-3/4})$. No step required weights to be nonnegative. $\square$

8.1 A computable violating hyperplane

Given a proposed reweighting, compute

$$KH = (1 - 1/r)H + U(U^T H)/r. \quad (8.9)$$

Choose a row i of largest norm. If it is nonzero, put $x = u_i$ and $y = (KH)_{i,:} / \|(KH)_{i,:}\|$. Among the four normals

$$(x, y), \quad (x, -y), \quad (2x, y), \quad (2x, -y)$$

one has relative error at least $\delta \|(KH)_{i,:}\|/10$. To see this, repeat (8.8) with the largest of the four relative errors in place of ε . Normalize the selected normal and take its orthogonal hyperplane. Homogeneity leaves its relative error unchanged. The implementation uses (8.9), avoiding an $M \times M$ matrix in the witness calculation.

9 Computational checks, provenance, and proof audit

The accompanying code ran successfully with 2,851 assertions. Exact rational arithmetic checks all stencil moments for orders 1 through 16, and exact extraction from signed sums of even powers $p = 4, 6, 8, 10, 12$. One-hundred-digit arithmetic checks the global Taylor remainder at 390 scalar points covering exponents just above two, nonintegers, odd integers, and even integers. These are tests of formulas and implementation; Lemma 5.2 proves the remainder for all scalar inputs.

The program constructs the finite fields and the real bases from scratch for $r = 4, 16, 64$. It checks every difference of binary alternating forms, every basis-pair Gram block, the cubic kernel identity, the tight-frame identity, and 20 integer-vector fourth-moment identities per dimension using integer arithmetic. It then exercises the four-query witness on exactly half-supported nonnegative weights. The recorded results are:

r	Tested support	Target ε_r	Witness relative error
4	24	0.0189959	0.669495
16	1152	0.00721688	0.265564
64	67584	0.00260766	0.107057

Concrete matrices, weights, and normalized violating hyperplanes for $r = 4, 16$ are saved as NumPy archives. The theorem, not these particular candidate weights, excludes *every* support below its lower bound.

Additional floating-point diagnostics test selected rectangular derivative matrices at $p = 2.5, 3, 3.5, 4, 5$, and 6, the product-cost derivative identity, its finite-difference remainder, hyperplane projection costs, and zero-padding. A deliberately separate $p = 3$ circular example has a negative derivative-kernel eigenvalue, approximately -0.179580 , showing why a blanket positive-semidefiniteness assertion would be wrong. All 2,324 subsets of size at most three of one 24-vector noise instance were tested numerically.

Important limitation of the finite tests. The small random-core examples do not satisfy the conservative Frobenius event used in Lemma 3.1; the results file explicitly reports this. Their selected singular values are measured diagnostics, not realizations certified under every hypothesis of the asymptotic lemma. Pivoted QR is used only as a heuristic for these examples, not as a claimed implementation of Theorem 2.2. The full large- r existence argument is analytic. No numerical test certifies a continuum of subspace queries, all real p , or the external restricted-invertibility theorem. The earlier quartic manuscript [6] is included unchanged in `prior_work/`. The new all-exponent theorem does not invoke any unproved claim from it. The polynomial extraction in the quartic argument is replaced here by a genuine remainder estimate for non-even powers, and the core is obtained by a separate random construction and column-selection theorem.

The proof audit records the following distinctions: entrywise kernels versus ordinary matrix powers; Frobenius closeness versus operator-norm closeness; legal query directions versus selected input rows; exact polynomial extraction versus noninteger remainders; positive weights versus signed weights; support count versus total weight; and proved existence versus finite numerical checks. No claim of published priority, peer review, or formal verification is made.

A Gaussian comparison with the needed dimension bound

We first give the finite Gaussian comparison used in Lemma 2.3. Let (X_i) and (Y_i) be centered Gaussian vectors over a finite index set such that

$$\mathbb{E}(X_i - X_j)^2 \leq \mathbb{E}(Y_i - Y_j)^2 \quad (i, j).$$

Then $\mathbb{E} \max_i X_i \leq \mathbb{E} \max_i Y_i$. To prove this, use the smooth maximum $F_\tau(z) = \tau^{-1} \log \sum_i e^{\tau z_i}$ and interpolate between independent copies of X and Y via $Z_s = \sqrt{1-s}X + \sqrt{s}Y$. If $a_i = e^{\tau Z_{s,i}} / \sum_j e^{\tau Z_{s,j}}$, Gaussian integration by parts gives

$$\frac{d}{ds} \mathbb{E} F_\tau(Z_s) = \frac{\tau}{4} \mathbb{E} \sum_{i,j} a_i a_j \left[\mathbb{E}(Y_i - Y_j)^2 - \mathbb{E}(X_i - X_j)^2 \right] \geq 0.$$

The integration-by-parts identity follows by integrating the Gaussian density derivative; a small independent Gaussian perturbation handles a singular covariance matrix by continuity. Let $\tau \rightarrow \infty$, using $\max_i z_i \leq F_\tau(z) \leq \max_i z_i + (\log m)/\tau$ for m indices.

Now put $p' = p/(p-1) < 2$. Index Gaussian processes by x on the Euclidean unit sphere in $\mathbb{R}^r$ and y in the unit $\ell_{p'}$ ball in $\mathbb{R}^L$:

$$X_{x,y} = \langle Gx, y \rangle, \quad Y_{x,y} = \langle g, x \rangle + \langle h, y \rangle,$$

where $g \in \mathbb{R}^r, h \in \mathbb{R}^L$ are independent standard Gaussians. For two indices let $a = \langle x, x' \rangle$ and $b = \langle y, y' \rangle$. The norm inequality $\|y\|_2 \leq \|y\|_{p'} \leq 1$ ensures $a, b \leq 1$. Direct calculation gives

$$\mathbb{E}(Y_{x,y} - Y_{x',y'})^2 - \mathbb{E}(X_{x,y} - X_{x',y'})^2 = 2(1-a)(1-b) \geq 0.$$

Apply the finite comparison to successively finer finite nets of the two compact index sets. The processes are continuous and their suprema are bounded by integrable finite-dimensional Gaussian norms, so passage to the full sets is valid. Duality then gives

$$\begin{aligned} \mathbb{E} \|G\|_{2 \rightarrow p} &= \mathbb{E} \sup_{x,y} X_{x,y} \leq \mathbb{E} \sup_{x,y} Y_{x,y} \\ &= \mathbb{E} \|g\|_2 + \mathbb{E} \|h\|_p \leq \sqrt{r} + (L\gamma_p)^{1/p}. \end{aligned}$$

This proves Lemma 2.3.

B Elementary concentration and finite-net facts

B.1 Nets and singular values

For $0 < \eta < 1$, a maximal η -separated subset of the Euclidean unit sphere in $\mathbb{R}^d$ is an η -net. Balls of radius $\eta/2$ around its points are disjoint and lie in the radius- $1 + \eta/2$ ball. Comparing volumes gives cardinality at most $(1 + 2/\eta)^d$.

For a linear map T , let $L_T = \|T\|_{\text{op}}$. If all unit net points have norm of their image at most b , then $L_T \leq b + \eta L_T$, so $L_T \leq b/(1 - \eta)$. If their images also have norm at least a , every unit vector has image norm at least $a - \eta L_T$. The same upper-net argument applies to any continuous seminorm. These are the net-extension facts used in Section 4.

B.2 Bernstein's bound

If independent centered variables Y_i satisfy $|Y_i| \leq 1$ and $V = \sum_i \mathbb{E}Y_i^2$, then for $t > 0$,

$$\mathbb{P}\left\{\sum_i Y_i \geq t\right\} \leq \exp\left(-\frac{t^2}{2(V + t/3)}\right). \quad (\text{B.1})$$

For $0 < \lambda < 3$, expand the exponential. Since $|\mathbb{E}Y_i^j| \leq \mathbb{E}Y_i^2$ for $j \geq 2$, and $j! \geq 2 \cdot 3^{j-2}$,

$$\mathbb{E}e^{\lambda Y_i} \leq 1 + \frac{\lambda^2 \mathbb{E}Y_i^2}{2(1 - \lambda/3)} \leq \exp\left(\frac{\lambda^2 \mathbb{E}Y_i^2}{2(1 - \lambda/3)}\right).$$

Multiply these bounds and use exponential Markov. For $V > 0$, choose $\lambda = t/(V + t/3)$ to obtain (B.1). If $V = 0$, all variables vanish almost surely and the event has probability zero. For $X_i \in [0, 1]$, apply the result to $Y_i = X_i - \mathbb{E}X_i$.

B.3 Gaussian norm tails

If X is chi-square with r degrees of freedom, its moment generating function is $\mathbb{E}e^{tX} = (1 - 2t)^{-r/2}$ for $t < 1/2$. Optimizing exponential Markov gives, for $a > 1$ or $0 < a < 1$, the corresponding upper or lower tail bound

$$\mathbb{P}\{X \geq ar\} \text{ or } \mathbb{P}\{X \leq ar\} \leq \exp\left[-\frac{r}{2}(a - 1 - \log a)\right].$$

At $a = 1/4$ and $a = 9/4$, both exponents are larger than $r/16$. Hence $g \sim N(0, I_r/r)$ satisfies $\mathbb{P}\{\|g\| \notin [1/2, 3/2]\} \leq 2e^{-r/16}$. For a standard Gaussian vector, the lower tail at norm $\sqrt{r}/2$ is at most $e^{-r/8}$, a still weaker valid bound. These estimates justify the constants used in the core and noise lemmas.

C Explicit complete real mutually unbiased bases

Let $r = 4^h = 2^{m+1}$, where $m = 2h - 1$ is odd. Work in $E = \mathbb{F}_{2^m}$ with trace $T : E \rightarrow \mathbb{F}_2$. Thus

$$T(z) = z + z^2 + \cdots + z^{2^{m-1}}, \quad T(z^2) = T(z), \quad T(1) = 1.$$

Let $V = \mathbb{F}_2 \oplus E$, an $(m+1)$ -dimensional binary vector space. For each $a \in E$, define

$$\beta_a((t, x), (s, y)) = T(a^2 xy) + T(ax)T(ay) + tT(ay) + sT(ax). \quad (\text{C.1})$$

This is bilinear, symmetric, and alternating. On the diagonal, the first two terms cancel because $T((ax)^2) = T(ax) = T(ax)^2$, and the last two cancel.

Lemma C.1. *For distinct $a, b \in E$, the alternating form $\beta_a + \beta_b$ is nondegenerate.*

Proof. Put $c = a + b \neq 0$, $d = a/c$, $X = cx$, and $Y = cy$. Expanding in characteristic two gives

$$\begin{aligned} (\beta_a + \beta_b)((t, x), (s, y)) &= T(XY) + T(dX)T(Y) + T(X)T(dY) \\ &\quad + T(X)T(Y) + tT(Y) + sT(X). \end{aligned}$$

If (t, x) is in the radical, varying s first gives $T(X) = 0$. The remaining condition for every Y becomes

$$T([X + T(dX) + t]Y) = 0.$$

The trace pairing is nondegenerate: for any nonzero Z , choose $Y = Z^{-1}$ to get $T(ZY) = T(1) = 1$. Hence $X = T(dX) + t \in \mathbb{F}_2$. As m is odd, the restriction of T to $\mathbb{F}_2$ is the identity. Thus $T(X) = 0$ implies $X = 0$, and then $t = 0$. Since $c \neq 0$, also $x = 0$. $\square$

Choose a binary basis $e_1, \dots, e_{m+1}$ of V . For coordinates z_i , put

$$q_a(z) = \sum_{i < j} \beta_a(e_i, e_j) z_i z_j \in \mathbb{F}_2.$$

Its polar form is β_a : $q_a(z + h) + q_a(z) + q_a(h) = \beta_a(z, h)$. For every linear functional $\ell : V \rightarrow \mathbb{F}_2$, form the real vector indexed by $z \in V$,

$$u_{a,\ell}(z) = r^{-1/2} (-1)^{q_a(z) + \ell(z)}. \quad (\text{C.2})$$

For fixed a , these r vectors form an orthonormal basis: multiplying binary characters by a fixed sign vector preserves their orthogonality.

For distinct a, b , an inner product is r^{-1} times a quadratic character sum $S = \sum_z (-1)^{q(z) + \ell(z)}$ with nondegenerate polar form β . Its square is

$$S^2 = \sum_{h \in V} (-1)^{q(h) + \ell(h)} \sum_{z \in V} (-1)^{\beta(z, h)} = r.$$

For $h \neq 0$, nondegeneracy makes the inner linear character nontrivial, so its sum is zero. For $h = 0$, its sum is r . Therefore all cross-basis inner products have absolute value $r^{-1/2}$. Each coordinate of every vector in (C.2) has that same absolute value, so the standard basis is unbiased to all of them. The $|E| = r/2$ constructed bases, plus the standard basis, give $B = r/2 + 1$ bases as required.

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